Benchmarking Class-Imbalance Mitigation Across Four Histopathology Datasets

Yannik Queisler

July 12, 2026

Abstract

Class imbalance in computational pathology depends on the input regime: labeled image patches support ordinary classification, whereas whole-slide image (WSI) diagnosis is commonly modeled as weakly supervised learning over patch-feature bags. This report benchmarks representative class-imbalance mitigation methods on four histopathology datasets—TCGA-UT, BRACS, CAMELYON16, and PANDA—chosen to span the difficulty–imbalance plane and to vary organ, label type, and annotation form. All benchmarks fix a frozen Virchow2 encoder (Vorontsov et al., 2024; Zimmermann et al., 2024), a bounded per-slide evidence budget, and patient-disjoint splits, so that comparisons within each regime isolate the imbalance mechanism. On TCGA-UT (32-class, head-to-tail ≈28:1), center-focused affinity loss (CFAL) reaches the highest patch-level mean macro F1; OKO set learning reaches the highest balanced accuracy and the best raw NLL and Brier among patch methods; and RankMix has the highest WSI-bag balanced accuracy while MDE-MIL has the highest WSI-bag macro F1. The three additional datasets test whether these results generalize: on BRACS (mild native imbalance ≈2.3:1, hard task) only OKO and CFAL visibly separate from the CE baseline; on CAMELYON16 (severe patch imbalance ≈39:1, separable task) every method reaches near-ceiling and the unweighted baseline is competitive; and on PANDA (mild imbalance, two targets of very different difficulty) the no-mitigation baseline is competitive in both regimes. Together, the four datasets show that the headroom for imbalance mitigation is governed by class separability in the frozen representation, not by the head-to-tail ratio.

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1 Introduction

Class imbalance changes what a pathology classifier is rewarded for learning. When common tumor types dominate training data, high aggregate accuracy can coexist with unreliable rare-class recall, especially if evaluation emphasizes prevalence-sensitive metrics. Long-tailed learning and medical-image evaluation literature therefore motivate classwise recall, macro F1, balanced accuracy, and calibration-aware probability metrics as primary outcomes in imbalanced settings (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

Imbalance methods often assume a specific prediction unit. Image-level methods use labeled images, image embeddings, or directly generated image samples, whereas WSI-level methods use weak labels over bags of instances together with attention, instance ranking, or bag-level representation learning. The broader computational-pathology pipeline these regimes sit in—whole-slide imaging, patching into tiles, and weakly supervised multiple-instance learning over frozen foundation-model features—is introduced in the companion datasets report (Queisler, 2026a). Full method definitions, imbalance mechanisms, and metric formulas are given in the companion methods report (Queisler, 2026b). This report uses both companions as fixed references and contributes only the experimental results.

The benchmark answers two nested questions. First, on TCGA-UT—a moderately imbalanced 32-class pan-cancer cohort and the richest analysis in the study—which imbalance mechanism helps the most in each regime when the encoder is already strong? Second, do those results generalize: does the same benchmark machinery produce the same ranking when pointed at three further datasets on their native distributions, spanning mild and severe imbalance, different organs, and different label types?

The four datasets span the difficulty–imbalance plane: TCGA-UT is the anchor with the deepest analysis; BRACS (mild imbalance, hard fine-grained subtyping task) and CAMELYON16 (severe imbalance, separable detection task) together test whether imbalance severity alone predicts difficulty or mitigation headroom; and PANDA provides a within-cohort control, using two mildly imbalanced targets of different separability on the same biopsies. This spread is the design: if imbalance ratio governed mitigation headroom, CAMELYON16 should show the largest gains; if class separability governs it, CAMELYON16 should show the smallest gains despite being the most imbalanced.

2 Benchmark Data and Setup

The clinical background, provenance, and full native class distributions of all four datasets are given in the companion datasets report (Queisler, 2026a). This section states only what the benchmark consumes.

2.1 Dataset Overview

All benchmarks use frozen Virchow2 2,560-dimensional features (concatenation of CLS token and mean patch token, cast to float32) and patient-disjoint (or slide-disjoint for CAMELYON16) train/validation/test splits across three random seeds (Vorontsov et al., 2024; Zimmermann et al., 2024).

2.2 Per-Dataset Benchmark Consumption

TCGA-UT. The dataset contains 1,608,060 patches from 8,736 diagnostic slides across 32 cancer types (Tomczak et al., 2015; Komura et al., 2022). Splits are assigned at the TCGA participant level (case-disjoint), using 5,025 training, 1,075 validation, and 1,075 test cases per seed; all slides from one participant remain in one partition. The patch benchmark caps each slide at 30 deterministic patches (empirical median of available tiles at resolution level 0),

Dataset	Target label(s)	Split unit	Evidence budget	Imbalance ratio
TCGA-UT	32 cancer types	Case (patient)	30 patches/slide	≈28:1
BRACS	7 breast-lesion subtypes	Patient	360 ROI tiles/slide (median)	≈2.3:1
CAMELYON16	Tumor / normal (patch); metastasis detection (WSI)	Slide	7874 tissue tiles/slide (median)	≈39:1 (patch)
PANDA	Cancer/benign (patch); ISUP 0–5 (WSI)	Biopsy	421 tissue tiles/biopsy (median)	≈4.7:1 (patch); ≈2.4:1 (WSI)

Table 1: Benchmark dataset overview. Evidence budget is the per-slide cap used in both regimes. Imbalance ratio is head-to-tail; see Queisler (2026a) for entropy-based $1 - H_{\text{norm}}$ values and full class distributions.

yielding a controlled manifest that is shared by all patch methods. The WSI-bag benchmark applies the same 30-instance deterministic cap to each slide's feature bag.

BRACS. Evidence is derived from the ROI set: each annotated ROI is tiled into 256×256 RGB patches, and each slide's pooled ROI tiles are capped at the empirical median (360 tiles per slide). This design preserves label fidelity because all instances originate from pathologist-selected lesion regions. Of 547 published WSIs, 387 contribute at least one ROI tile; the benchmark covers 151 patients and 99,352 extracted patches.

CAMELYON16. Binary labels are mask-derived: each tile is labelled tumor or normal by intersecting it with the published annotation mask. Of 399 slides, 398 have usable tile evidence; the 20 tumor slides whose annotations are not exhaustive are excluded from the patch regime but retained in the WSI-bag regime, where the reliable slide-level label is used. The per-slide evidence budget is the empirical median of available tissue tiles, which for CAMELYON16 is 7,874—two orders of magnitude larger than TCGA-UT's 30-tile cap, reflecting that a whole lymph-node section yields far more tissue than a single lesion ROI or tumor-region crop. Each slide is its own patient, so slide-disjoint splits coincide with patient-disjoint splits.

PANDA. The WSI-bag regime uses the canonical six-class ISUP grade (0 = benign, 1–5 = increasing cancer grade groups). The patch regime uses a provider-consistent binary cancer/benign target derived from pixel masks; because Radboud and Karolinska masks use institution-dependent value semantics, collapsing both to cancer/benign is the only label the two providers share. To keep frozen-feature extraction tractable, the benchmark uses a stratified subset of approximately 2,000 biopsies, sampled to preserve the joint distribution of ISUP grade and provider. The per-slide evidence budget is the empirical median number of tissue tiles per biopsy (421). Biopsies without available masks are excluded from the patch regime but kept in the WSI-bag regime.

2.3 Shared Protocol

All datasets use three random seeds with patient-disjoint splits, deterministic per-slide evidence caps, and frozen Virchow2 features. Hyperparameters are selected independently per method: for each candidate configuration the validation macro F1 is averaged over the three seeds, and the configuration with the highest mean is chosen (balanced accuracy as tie-breaker). A single configuration per method is fixed before evaluating on the held-out test split. Results are reported as mean ± standard deviation across seeds. Patch and WSI-bag methods are compared only within their own regime.

3 Experimental Setup

Full method definitions and metric formulas are given in the companion methods report (Queisler, 2026b). This section records only the configurations used to run the methods.

3.1 Method Roster

Family	Methods	Regime
No-mitigation baseline	CE, MIL CE	Patch, WSI bag
Data-level sampling and augmentation	Balanced sampling, RankMix	Patch, WSI bag
Algorithm-level losses	Weighted CE, Focal loss, CFAL	Patch (CFAL); Patch and WSI bag (others)
Hybrid sampling–loss	CE + soft F1, CE + soft MCC, OKO, MDE-MIL	Patch (first three, OKO); WSI bag (MDE-MIL)
Synthetic data generation	ProGAN augmentation	Patch
Representation and architecture	SC-MIL	WSI bag

Table 2: Method families and regimes. CE and MIL CE are unweighted cross-entropy baselines so mitigation gains are measured against a genuine no-mitigation reference. Patch and WSI-bag methods are compared only within their own input regime. Method definitions follow Queisler (2026b).

3.2 Training Protocol

Setting	Patch benchmark	WSI-bag benchmark
Optimizer	AdamW	AdamW
Learning rate	10^{-3}	10^{-3}
Weight decay	10^{-4}	10^{-4}
Epochs	30	30
Early stopping	none	none
Minibatch size	256 patches	32 slide bags
Model selection	final epoch	final epoch
Random seeds	0, 1, 2	0, 1, 2

Table 3: Shared training protocol for both benchmarks. All methods within a benchmark use the same optimizer, schedule, and epoch budget unless noted in the method documentation (Queisler, 2026b).

3.3 Tuning Grids

3.4 Classifier Heads

Figure 1 shows the patch and WSI-bag classifier heads shared across all datasets.
The patch head is a two-layer MLP: linear map to 512 units, ReLU, dropout rate 0.1, and a linear output to C classes. CFAL keeps the same trunk but replaces the final linear map with one learnable prototype per class and Gaussian affinity-based prediction (Queisler, 2026b). The

Regime	Method	Reference default	Validation grid
Patch	Weighted CE	Weight power = 1	$\{0.25, 0.5, 0.75, 1\}$
WSI bag	Weighted CE	Weight power = 1	$\{0, 0.125, 0.25, 0.5, 0.75, 1\}$
Patch, WSI	Balanced sampler	Sampler power = 1	$\{0.5, 0.75, 1\}$
Patch, WSI	Focal loss	$\gamma = 2$	$\gamma \in \{0.5, 1, 1.5, 2\}$
Patch	CE + soft F1	Metric-loss weight = 1	$\{0.25, 0.5, 1, 2, 4, 8, 16, 32, 64, 256, 512, 1024\}$
Patch	CE + soft MCC	Metric-loss weight = 1	$\{0.25, 0.5, 1, 2, 8, 32\}$
Patch	CFAL	$\gamma = 2$, $\sigma = 1$	$\sigma \in \{0.25, 1, 4\}$
Patch	ProGAN augmentation	Final-depth epochs = 25	$\{10, 25, 50\}$
Patch	OKO	$k = 1$	$k \in \{1, 2, 3, 4, 5, 6, 8, 12, 20\}$
WSI bag	RankMix	$\alpha = 1$	$\alpha \in \{0.2, 0.5, 1, 2, 4, 8, 16, 32, 128\}$
WSI bag	SC-MIL	$\tau = 1$	$\tau \in \{0.05, 0.1, 0.2, 0.5\}$
WSI bag	MDE-MIL	$\lambda_{\text{con}} = 0.25$	$\lambda_{\text{con}} \in \{0.1, 0.25, 0.3, 0.5, 1, 2, 4\}$

Table 4: Method-specific imbalance controls. Each row varies one control on validation while retaining the shared budget in Table 3. ProGAN sweeps only the final-depth refinement epochs; generator training follows Table 5. Identical grids are applied on all four datasets.

Setting	Value
Resolution stages	4×4 to 256×256 (seven stages)
Epochs per stage (depths 1–6)	25
Final-depth (256×256) refinement epochs	25 (reference); validation selects from $\{10, 25, 50\}$
Fade-in fraction for new layers	0.5 (pinned to the reference 25-epoch schedule)
Latent dimension	128
Base channel width	512
Adversarial loss	WGAN-GP ($\lambda = 10$, drift $= 10^{-3}$)
Equalized learning rate	Yes (weight $\sim \mathcal{N}(0, 1)$, scale $\sqrt{2/\text{fan\_in}}$ at forward)
Optimizer	Adam, learning rate 2×10^{-4}, $\beta_1 = 0.5$
Real patches per class for GAN training	Up to 2,048, deterministic per-class subsample
Synthetic count per tail class	Head-class train count minus class train count

Table 5: ProGAN generator training settings. Depths 1–6 use 25 epochs each. For the final depth, one long run trains to 50 epochs and the generator is snapshotted at epochs 10, 25, and 50; validation selects the snapshot with the best mean macro F1 over seeds, reusing one GAN training run for all three sweep points. Evaluated on TCGA-UT only (patch regime).

WSI-bag head is an attention-based MIL network: each instance passes through the shared encoder ($2560 \to 256$), instance weights come from a linear scorer followed by softmax over instances, the bag representation is their weighted sum, and slide logits are produced by a final linear map. SC-MIL adds a two-layer projection head of width 256 with ReLU. MDE-MIL keeps the attention pooling but uses two separate expert heads. The WSI-bag input is capped at the dataset-specific evidence budget (Table 1).

3.5 Evaluation

Macro F1 is the primary ranking metric within each regime because it weights classes equally and penalizes failure on rare classes. Balanced accuracy and accuracy are reported alongside it. For TCGA-UT, head/body/tail class-group metrics group the 32 cancer types into fixed support tiers from the full-dataset slide distribution (8 tail, 16 body, 8 head classes). Three complementary probability-quality metrics—negative log likelihood (NLL), multiclass Brier score, and expected calibration error (ECE)—are reported on held-out softmax outputs, both before and after a unified post-hoc temperature-scaling protocol. For each method and seed, a single temperature is fit on validation by minimizing NLL and applied unchanged to the test score matrix.

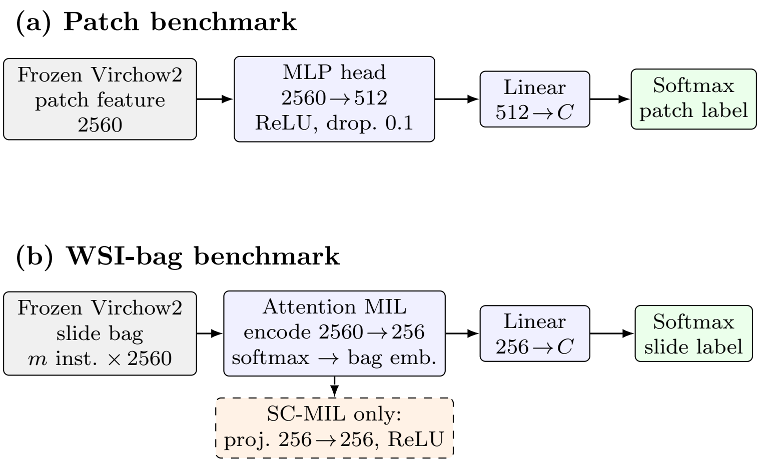

Figure 1: Shared classifier heads for the benchmark (C = number of classes in each dataset). Panel (a) classifies one controlled patch embedding per forward pass. CFAL reuses the MLP trunk but replaces the linear $512 \to C$ head with learnable prototypes and affinity-based prediction. Panel (b) encodes all instances on a slide, pools them with attention, and predicts one slide-level label; the dashed branch is used only by SC-MIL. MDE-MIL reuses the same attention trunk but replaces the single linear head with two expert heads. Virchow2 features are frozen in both benchmarks.

Full metric definitions and the temperature-scaling protocol are given in Queisler (2026b). Patch and WSI-bag methods are never ranked against each other.

4 Results

4.1 TCGA-UT

Both TCGA-UT benchmarks share case-disjoint splits, the Virchow2 feature family, and a maximum of 30 feature instances per slide, but they answer different prediction questions. The patch benchmark treats each selected crop as an independent labeled example; the WSI-bag benchmark predicts one slide label from an attention-weighted aggregation. Absolute scores and method orderings are not comparable across the two tables. Supporting diagnostic tables and figures are in Appendix A.

4.1.1 Selected Controls and Tuning Sensitivity

How much the selected control matters varies sharply across methods. A seed-blocked repeated-measures analysis of the validation sweeps shows that across-configuration variation exceeds twice the seed standard deviation only for OKO (range-to-seed-noise ratio 75.8) and RankMix (ratio 2.06, $p < 0.01$). For all single-knob resampling, reweighting, and focal mechanisms on frozen Virchow2 embeddings the swept control produces variation within seed noise regardless of its value.

4.1.2 Patch-Level Results

The patch benchmark compresses into a narrow performance band once patches are represented with frozen Virchow2 embeddings (Table 7). CFAL ranks highest on macro F1, while OKO ranks highest on balanced accuracy. CFAL has paired gains over plain CE on both headline metrics in all three seeds (Table 25). OKO edges CFAL on balanced accuracy and achieves macro F1 between CE plus soft F1 and CE plus soft MCC; CE plus soft F1 and OKO both improve on CE

Regime	Method	Selected control	Val macro F1	Test macro F1	Test bal. acc.
Patch	CE	baseline (no sweep)	–	0.773	0.764
Patch	Balanced sampler	sampler_power=1	0.768	0.771	0.769
Patch	CE + soft F1 (balanced)	metric_loss_weight=256	0.781	0.780	0.781
Patch	CE + soft MCC (balanced)	metric_loss_weight=32	0.771	0.770	0.772
Patch	CFAL	cfal_sigma=1	0.792	0.798	0.789
Patch	Focal	focal_gamma=1.5	0.764	0.765	0.758
Patch	OKO	oko_k=4	0.783	0.788	0.791
Patch	ProGAN augmentation	final_depth_epochs=10	0.756	0.766	0.760
Patch	Weighted CE	weight_power=0.5	0.764	0.774	0.767
WSI bag	MIL CE	baseline (no sweep)	–	0.826	0.823
WSI bag	MDE-MIL (ensemble)	mde_mil_consistency_weight=2	0.851	0.865	0.849
WSI bag	Balanced MIL	sampler_power=0.75	0.821	0.831	0.829
WSI bag	Focal MIL	focal_gamma=0.5	0.823	0.824	0.823
WSI bag	Weighted MIL	weight_power=0.125	0.828	0.835	0.833
WSI bag	RankMix	rankmix_alpha=32	0.843	0.854	0.864
WSI bag	SC-MIL	sc_mil_temperature=0.1	0.828	0.839	0.837

Table 6: Validation-selected TCGA-UT configuration provenance (Section 3.2). Configurations are chosen by mean validation macro F1 over seeds 0–2 with balanced accuracy as tie-breaker; held-out test metrics in Tables 7 and 9 are reported after selection.

Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.847 ± 0.008	0.789 ± 0.007	0.798 ± 0.009
OKO	0.841 ± 0.006	0.791 ± 0.005	0.788 ± 0.003
CE + soft F1 (balanced)	0.835 ± 0.004	0.781 ± 0.002	0.780 ± 0.003
Weighted CE	0.826 ± 0.008	0.767 ± 0.006	0.774 ± 0.005
CE	0.828 ± 0.010	0.764 ± 0.011	0.773 ± 0.015
Balanced sampler	0.827 ± 0.010	0.769 ± 0.007	0.771 ± 0.014
CE + soft MCC (balanced)	0.824 ± 0.009	0.772 ± 0.004	0.770 ± 0.005
ProGAN augmentation	0.824 ± 0.004	0.760 ± 0.005	0.766 ± 0.002
Focal	0.825 ± 0.007	0.758 ± 0.007	0.765 ± 0.009

Table 7: TCGA-UT validation-selected patch-level test results over three seeds. Methods share the same controlled manifest, frozen Virchow2 encoder, shared MLP trunk, and evaluation code.

in the paired comparison, but absolute gaps remain modest and standard deviations overlap. Weighted CE, balanced sampling, and focal loss remain within the broader CE-centered band.

ProGAN augmentation stays within the non-generative spread despite adding a large synthetic tail-class volume (Table 8). One plausible explanation is the frozen-feature pipeline: the class-specific generators reach an Inception FID of 239.5–382.0 (median 293.4), comparable to the ProGAN FID reported by Ruiz-Casado et al. (2024) on a binary breast-histology task where GAN augmentation produced large downstream gains using an end-to-end convolutional classifier. This benchmark instead embeds every patch with a frozen Virchow2 encoder; a frozen encoder may map synthetic patches into embedding regions the head rarely sees for real data, limiting synthetic-volume gains even when pixel-level FID is reasonable. This frozen-feature explanation remains a hypothesis pending feature-space diagnostics such as real-to-real nearest-neighbor baselines or maximum mean discrepancy tests.

Class	Real train	Generated	Inception FID	Virchow2 mean NN
Lymphoid Neoplasm Diffuse Large B-cell Lymphoma	600	16,240	345.6	50.68
Cholangiocarcinoma	660	16,180	307.6	51.79
Pheochromocytoma and Paraganglioma	900	15,940	382.0	54.06
Uveal Melanoma	1,160	15,680	338.4	48.88
Rectum adenocarcinoma	1,350	15,490	239.5	47.97
Mesothelioma	1,430	15,410	286.3	51.23
Uterine Carcinosarcoma	1,550	15,290	304.0	50.94
Kidney Chromophobe	1,740	15,100	282.8	48.97
All augmented classes (31 total)	165,470	356,570	293.4 median	51.32 median

Table 8: ProGAN augmentation diagnostics for seed 0. "Generated" counts are synthetic patches added per class to reach the head-class training patch count. Inception FID compares generated patches to the real training subset. Virchow2 mean NN is the mean, over a fixed synthetic subsample, of the minimum L_2 distance to real same-class embeddings in the frozen feature cache.

Method	Accuracy	Balanced accuracy	Macro F1
MDE-MIL (ensemble)	0.896 ± 0.002	0.849 ± 0.014	0.865 ± 0.011
RankMix	0.890 ± 0.005	0.864 ± 0.002	0.854 ± 0.008
SC-MIL	0.883 ± 0.013	0.837 ± 0.013	0.839 ± 0.008
Weighted MIL	0.882 ± 0.006	0.833 ± 0.019	0.835 ± 0.011
Balanced MIL	0.875 ± 0.003	0.829 ± 0.010	0.831 ± 0.009
MIL CE	0.873 ± 0.013	0.823 ± 0.014	0.826 ± 0.013
Focal MIL	0.872 ± 0.006	0.823 ± 0.019	0.824 ± 0.008

Table 9: TCGA-UT validation-selected WSI-bag test results over three seeds. Methods share the same capped feature bags, attention-MIL backbone family, and evaluation code.

4.1.3 WSI-Bag Results

The WSI-bag benchmark evaluates one slide-level prediction per diagnostic slide by aggregating capped 30-instance feature bags. RankMix is above plain MIL CE on both headline metrics in all three paired seeds (Table 25) and has the highest mean balanced accuracy, although its macro-F1 gain is more modest (Figure 5). MDE-MIL reaches the highest mean test macro F1, with positive paired deltas in all three seeds but wider absolute spread. On raw probability quality, MDE-MIL has the lowest mean NLL, Brier, and ECE among WSI methods, with RankMix second; both clearly improve on plain MIL CE before temperature scaling.

Other WSI methods cluster closely below this pair. Focal MIL is unstable across seeds: competitive on seeds 0 and 2 (macro F1 0.846 and 0.845) but collapsing on seed 1 (0.811), giving the widest seed-to-seed standard deviation (0.020) in the WSI benchmark.

4.1.4 Paired Seed Comparisons

Table 25 makes the headline gains explicit at seed resolution. CFAL is above plain CE on macro F1 and balanced accuracy in every seed; OKO is above CE on both metrics in every seed; CE plus soft F1 is above CE on balanced accuracy in every seed and on macro F1 in two of three. On WSI bags, both RankMix and MDE-MIL are above plain MIL CE on both headline metrics in every seed. The absolute shifts are modest and would require formal paired testing to treat as significant.

4.1.5 Probability Calibration

Table 26 reports raw probability-quality metrics. CFAL improves discrimination by the largest margin among patch methods but has the highest raw ECE and worst NLL/Brier scores—a typical over-confidence pattern when geometric affinities are passed through $\log a_{ic}$ and softmax. OKO is the notable exception to the usual discrimination–calibration tradeoff: it achieves the best raw NLL and Brier among patch methods while also placing highest on balanced accuracy. Focal loss attains the smallest raw ECE, so no patch method dominates all three raw calibration metrics. On WSI bags, MDE-MIL reaches the lowest raw NLL, Brier, and ECE, with RankMix second; both clearly improve on plain MIL CE before temperature scaling, while the other MIL variants cluster near the MIL CE baseline.

Regime	Method	NLL	Brier	ECE
Patch	CFAL	0.532 ± 0.029	0.220 ± 0.011	0.019 ± 0.003
Patch	OKO	0.617 ± 0.032	0.236 ± 0.011	0.029 ± 0.006
Patch	CE + soft F1 (balanced)	0.780 ± 0.026	0.263 ± 0.008	0.061 ± 0.028
Patch	Weighted CE	0.654 ± 0.030	0.252 ± 0.012	0.038 ± 0.005
Patch	CE	0.649 ± 0.035	0.250 ± 0.014	0.041 ± 0.002
Patch	Balanced sampler	0.653 ± 0.034	0.252 ± 0.013	0.038 ± 0.008
Patch	CE + soft MCC (balanced)	0.768 ± 0.034	0.271 ± 0.013	0.076 ± 0.029
Patch	ProGAN augmentation	0.723 ± 0.017	0.263 ± 0.005	0.073 ± 0.007
Patch	Focal	0.618 ± 0.025	0.251 ± 0.009	0.006 ± 0.004
WSI bag	MDE-MIL (ensemble)	0.471 ± 0.009	0.163 ± 0.003	0.023 ± 0.002
WSI bag	RankMix	0.428 ± 0.012	0.166 ± 0.005	0.028 ± 0.002
WSI bag	SC-MIL	0.440 ± 0.045	0.173 ± 0.018	0.007 ± 0.004
WSI bag	Weighted MIL	0.455 ± 0.033	0.177 ± 0.011	0.013 ± 0.004
WSI bag	Balanced MIL	0.479 ± 0.031	0.185 ± 0.010	0.013 ± 0.003
WSI bag	MIL CE	0.473 ± 0.018	0.188 ± 0.015	0.017 ± 0.010
WSI bag	Focal MIL	0.475 ± 0.024	0.189 ± 0.008	0.012 ± 0.002

Table 10: TCGA-UT held-out test calibration metrics after method-agnostic post-hoc temperature scaling. For each method and seed, the temperature is fit only on that seed's validation split and then applied to the test split. Table 26 (Appendix) gives the raw-score diagnostic.

After temperature scaling, CFAL reaches the best mean NLL and Brier score—consistent with treating affinity outputs as rank scores that need a monotone probability map—while focal loss reaches the lowest mean post-hoc ECE. On WSI bags, RankMix has the lowest mean post-hoc NLL and MDE-MIL the lowest mean post-hoc Brier; after scaling, however, the simpler MIL baselines reach lower mean ECE than RankMix or MDE-MIL, so the raw ECE advantage should be read partly as a confidence-scale mismatch.

4.1.6 Classwise Difficulty and Support

Table 11 and Figures 7–8 make the support–difficulty mismatch explicit. Rectum adenocarcinoma is the hardest class in both regimes despite being the fifth-lowest-support class rather than the rarest. Conversely, the rarest class, diffuse large B-cell lymphoma, does not appear among the five hardest classes in either regime. The rank association between dataset slide support and CE-baseline recall is positive but modest: Spearman $\rho = 0.334$ for patches and $\rho = 0.301$ for WSI bags.

4.2 BRACS

The patch-feature and WSI-bag regimes are reported separately (Tables 12 and 13). Absolute scores are markedly lower than on TCGA-UT for two concrete reasons: fine-grained breast-lesion

Regime	Class	Slides	Support rank	Difficulty rank	Recall	F1
Patch	Rectum adenocarcinoma	63	5	1	0.103	0.144
Patch	Cholangiocarcinoma	30	2	2	0.319	0.409
Patch	Esophageal carcinoma	113	10	3	0.362	0.420
Patch	Uterine Carcinosarcoma	71	7	4	0.539	0.519
Patch	Mesothelioma	70	6	5	0.556	0.628
WSI-bag	Rectum adenocarcinoma	63	5	1	0.338	0.409
WSI-bag	Esophageal carcinoma	113	10	2	0.523	0.564
WSI-bag	Cholangiocarcinoma	30	2	3	0.528	0.544
WSI-bag	Uterine Carcinosarcoma	71	7	4	0.625	0.611
WSI-bag	Stomach adenocarcinoma	274	18	5	0.693	0.735

Table 11: Hardest TCGA-UT classes under each regime's CE baseline, averaged over seeds 0–2 on the held-out test split. Support rank is computed from full TCGA-UT slide counts (rank 1 = lowest support). Difficulty rank orders classes by baseline error rate, 1 − recall.

subtyping is intrinsically harder than pan-cancer type classification, and the native breast-only training pool is far smaller. Even recent task-specific BRACS models report only mid-60% weighted F1 (Pati et al., 2022; Stegmüller et al., 2023); a frozen-feature benchmark under a bounded evidence budget should not approach TCGA-UT-level discrimination.

Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.589 ± 0.027	0.497 ± 0.042	0.490 ± 0.038
OKO	0.565 ± 0.054	0.502 ± 0.052	0.488 ± 0.049
CE + soft F1	0.547 ± 0.034	0.480 ± 0.036	0.468 ± 0.033
CE + soft MCC	0.545 ± 0.043	0.472 ± 0.047	0.463 ± 0.044
Balanced sampling	0.552 ± 0.045	0.470 ± 0.043	0.461 ± 0.045
ProGAN augmentation	0.549 ± 0.049	0.460 ± 0.045	0.457 ± 0.049
Focal loss	0.533 ± 0.041	0.460 ± 0.037	0.452 ± 0.036
CE	0.548 ± 0.045	0.451 ± 0.056	0.451 ± 0.055
Weighted CE	0.531 ± 0.045	0.461 ± 0.040	0.450 ± 0.042

Table 12: BRACS native patch-feature test results for validation-selected configurations, mean ± standard deviation across patient-disjoint seeds.

In the patch regime, OKO and CFAL are the only methods that visibly separate from the unweighted CE baseline, leading by roughly five balanced-accuracy points, while every other method—including ProGAN augmentation—falls inside one standard deviation of CE. In the WSI-bag regime, the unweighted MIL CE baseline has the highest balanced accuracy and macro F1, with no mitigation method improving on it—the failure mode the no-mitigation baseline is included to expose.

Probability calibration. In the patch regime, raw miscalibration is severe for most methods (ECE 0.29–0.42); OKO is the exception, with by far the lowest raw ECE (0.045) and a fitted temperature close to one ($T \approx 0.88$), echoing its behaviour on TCGA-UT. CFAL again shows the near-degenerate low-temperature signature ($T \approx 0.08$) that scaling nonetheless corrects. In the WSI-bag regime, calibration is uniformly worse than in the patch regime and than on TCGA-UT: post-scaling ECE ranges from 0.136 (focal loss) to 0.168 (balanced sampling and SC-MIL), with the elevated residual reflecting the smaller training pool and fine-grained label

Method	Accuracy	Balanced accuracy	Macro F1
MIL CE	0.672 ± 0.020	0.524 ± 0.103	0.508 ± 0.045
Weighted CE	0.667 ± 0.029	0.510 ± 0.093	0.501 ± 0.055
Focal loss	0.691 ± 0.029	0.490 ± 0.032	0.498 ± 0.033
MDE-MIL	0.635 ± 0.033	0.489 ± 0.093	0.480 ± 0.051
Balanced sampling	0.635 ± 0.022	0.449 ± 0.022	0.462 ± 0.023
RankMix	0.603 ± 0.104	0.437 ± 0.087	0.446 ± 0.075
SC-MIL	0.539 ± 0.092	0.367 ± 0.083	0.383 ± 0.086

Table 13: BRACS native WSI-bag test results for validation-selected configurations, mean ± standard deviation across patient-disjoint seeds.

Method	ECE	ECE+TS	T
CE	0.354 ± 0.039	0.031 ± 0.010	5.812 ± 0.409
Weighted CE	0.367 ± 0.039	0.038 ± 0.017	5.752 ± 0.486
Focal loss	0.286 ± 0.023	0.030 ± 0.017	3.459 ± 0.405
CFAL	0.417 ± 0.026	0.049 ± 0.013	0.079 ± 0.010
CE + soft F1	0.405 ± 0.030	0.070 ± 0.027	8.532 ± 1.329
CE + soft MCC	0.366 ± 0.039	0.029 ± 0.011	6.582 ± 0.723
Balanced sampling	0.347 ± 0.036	0.021 ± 0.011	5.823 ± 0.601
OKO	0.045 ± 0.028	0.039 ± 0.014	0.877 ± 0.149
ProGAN augmentation	0.358 ± 0.041	0.039 ± 0.018	6.158 ± 0.677

Table 14: BRACS native patch-feature calibration: ECE before (raw) and after (TS) temperature scaling, with the fitted temperature T, mean ± standard deviation across seeds.

set.

4.3 CAMELYON16

On CAMELYON16 the picture inverts: the same frozen features that struggle with BRACS's fine-grained subtypes separate tumor from normal tissue almost perfectly, so despite patch imbalance an order of magnitude more severe every method reaches near-ceiling discrimination in both regimes (Tables 16 and 17).

In the patch regime, balanced accuracy ranges from 0.961 to 0.986 and the unweighted CE baseline (BAcc 0.962) sits squarely inside the field. In the WSI-bag regime, all methods score between balanced accuracy 0.965 and 0.984; the unweighted MIL CE baseline is again fully competitive, and the entire field is separated by less than 0.02 macro F1. Canonical slide-level metastasis detection is nearly solved from bounded Virchow2 feature bags.

Probability calibration. Calibration follows discrimination: raw patch ECE is 0.003–0.005 for every method except CFAL, and temperature scaling brings it to 0.001–0.002. CAMELYON16 patch and WSI-bag calibration is markedly better than BRACS—patch post-scaling ECE around 0.001 versus BRACS's 0.04–0.08—which is the calibration counterpart of the same separability gap.

Method	ECE	ECE+TS	T
MIL CE	0.237 ± 0.012	0.165 ± 0.040	3.092 ± 0.322
Weighted CE	0.247 ± 0.051	0.163 ± 0.045	2.928 ± 0.343
Focal loss	0.164 ± 0.066	0.136 ± 0.010	1.977 ± 0.075
Balanced sampling	0.257 ± 0.064	0.168 ± 0.052	2.861 ± 0.314
RankMix	0.151 ± 0.074	0.144 ± 0.017	1.298 ± 0.108
SC-MIL	0.302 ± 0.097	0.168 ± 0.014	2.421 ± 0.242
MDE-MIL	0.277 ± 0.017	0.151 ± 0.010	2.083 ± 0.326

Table 15: BRACS native WSI-bag calibration: ECE before and after temperature scaling, with the fitted temperature T. Columns as in Table 14.

Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.997 ± 0.002	0.961 ± 0.015	0.966 ± 0.008
Balanced sampling	0.997 ± 0.001	0.969 ± 0.014	0.960 ± 0.020
ProGAN augmentation	0.996 ± 0.001	0.956 ± 0.009	0.959 ± 0.019
CE	0.996 ± 0.001	0.962 ± 0.015	0.958 ± 0.020
Focal loss	0.996 ± 0.001	0.972 ± 0.010	0.957 ± 0.027
OKO	0.996 ± 0.000	0.974 ± 0.007	0.955 ± 0.034
CE + soft F1	0.996 ± 0.001	0.986 ± 0.002	0.952 ± 0.035
Weighted CE	0.996 ± 0.002	0.966 ± 0.012	0.952 ± 0.028
CE + soft MCC	0.995 ± 0.001	0.982 ± 0.006	0.940 ± 0.050

Table 16: CAMELYON16 native patch-feature test results for validation-selected configurations, mean ± standard deviation across seeds.

4.4 PANDA

PANDA is reported in the same two regimes on its regime-specific labels: tile-level binary cancer detection in the patch regime (Table 20) and slide-level six-class ISUP grading in the WSI-bag regime (Table 21). The two targets are both mildly imbalanced but sit on opposite sides of the difficulty axis within one cohort: binary cancer/benign detection is coarse, while six-class ISUP grading from a bounded feature bag is a fine-grained ordinal problem on which even purpose-built end-to-end models trained for the PANDA challenge reach only moderate agreement (Bulten et al., 2022).

In the patch regime, every method reaches balanced accuracy between 0.832 and 0.860 and macro F1 between 0.811 and 0.866. The unweighted CE baseline sits squarely inside this band (BAcc 0.838, macro F1 0.859), and no mitigation method separates from it by more than between-seed noise.

The WSI-bag regime inverts the difficulty: macro F1 spans only 0.559–0.590 and balanced accuracy 0.565–0.588—roughly 0.27 below the detection regime on the very same biopsies. The unweighted MIL CE baseline (BAcc 0.566, F1 0.565) is within noise of the field; balanced sampling, focal loss, and MDE-MIL lead it by about 0.02, but between-seed standard deviations of 0.02–0.04 make these numerical rather than statistically separated leads.

Probability calibration. Calibration follows separability: patch regime raw ECE is 0.024–0.103 and post-scaling ECE is 0.005–0.036; OKO is the best calibrated (post-scaling ECE 0.005). In the harder WSI-bag regime, raw ECE is 0.129–0.252 and residual post-scaling ECE is 0.039–0.097—the calibration counterpart of the detection-versus-grading separability gap.

Method	Accuracy	Balanced accuracy	Macro F1
MDE-MIL	0.983 ± 0.017	0.984 ± 0.018	0.982 ± 0.018
SC-MIL	0.983 ± 0.017	0.984 ± 0.018	0.982 ± 0.018
Balanced sampling	0.977 ± 0.010	0.979 ± 0.012	0.977 ± 0.010
Focal loss	0.977 ± 0.010	0.979 ± 0.012	0.977 ± 0.010
RankMix	0.972 ± 0.020	0.974 ± 0.020	0.971 ± 0.020
MIL CE	0.966 ± 0.017	0.965 ± 0.021	0.965 ± 0.018
Weighted CE	0.966 ± 0.017	0.965 ± 0.021	0.965 ± 0.018

Table 17: CAMELYON16 native WSI-bag test results (slide-level metastasis detection) for validation-selected configurations, mean ± standard deviation across seeds.

Method	ECE	ECE+TS	T
CE	0.003 ± 0.001	0.001 ± 0.001	3.723 ± 0.142
Weighted CE	0.004 ± 0.002	0.001 ± 0.001	4.224 ± 0.503
Focal loss	0.003 ± 0.001	0.001 ± 0.000	4.547 ± 0.173
CFAL	0.464 ± 0.001	0.068 ± 0.004	0.050 ± 0.000
CE + soft F1	0.004 ± 0.001	0.001 ± 0.001	4.598 ± 0.179
CE + soft MCC	0.005 ± 0.001	0.002 ± 0.000	5.025 ± 0.191
Balanced sampling	0.003 ± 0.001	0.001 ± 0.001	4.071 ± 0.239
OKO	0.003 ± 0.000	0.001 ± 0.000	3.014 ± 0.100
ProGAN augmentation	0.003 ± 0.001	0.001 ± 0.001	4.119 ± 0.281

Table 18: CAMELYON16 native patch-feature calibration: ECE before (raw) and after (TS) temperature scaling, with the fitted temperature T, mean ± standard deviation across seeds.

4.5 Cross-Dataset Summary

Table 24 and Figure 2 collapse the full results to the comparison needed for the report's central claim. The largest positive deltas appear where the task is already hard and classes are entangled (BRACS patch, PANDA WSI-bag). Near-ceiling regimes (CAMELYON16 patch and WSI-bag) show tiny positive deltas because there is almost no error left to remove. Negative or zero deltas (BRACS WSI-bag, PANDA patch) indicate that the no-mitigation baseline is competitive, a pattern that the no-mitigation reference is specifically included to surface.

5 Discussion

(a) TCGA-UT lesson: frozen-encoder bottleneck. On TCGA-UT, once a strong frozen pathology encoder is fixed, patch-level gains come from reshaping the decision head or objective—CFAL through prototype-affinity training and OKO through set-aggregated logit regularization—rather than from resampling, difficulty reweighting, or synthetic-image augmentation, which barely move the CE baseline. CFAL replaces the linear head with learnable class prototypes and a margin objective, pulling examples toward compact class centers and penalizing confusion with nearby wrong-class prototypes. OKO constructs training sets with one matched pair and k distractors; averaging per-example logits across set members implicitly regularizes logit magnitude and suppresses the overconfidence that afflicts CE-trained classifiers, which explains its combination of high balanced accuracy and the best raw NLL and Brier among patch methods. The simpler reweighting and resampling variants (weighted CE, balanced sampling, focal loss)

Method	ECE	ECE+TS	T
MIL CE	0.027 ± 0.017	0.034 ± 0.017	0.050 ± 0.000
Weighted CE	0.027 ± 0.017	0.034 ± 0.017	0.050 ± 0.000
Focal loss	0.021 ± 0.013	0.023 ± 0.010	0.050 ± 0.000
Balanced sampling	0.022 ± 0.009	0.023 ± 0.010	0.050 ± 0.000
RankMix	0.022 ± 0.030	0.028 ± 0.024	0.273 ± 0.386
SC-MIL	0.020 ± 0.018	0.017 ± 0.017	0.050 ± 0.000
MDE-MIL	0.013 ± 0.008	0.017 ± 0.017	0.050 ± 0.000

Table 19: CAMELYON16 native WSI-bag calibration: ECE before and after temperature scaling, with the fitted temperature T. Columns as in Table 18.

Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.928 ± 0.004	0.832 ± 0.003	0.866 ± 0.002
CE	0.921 ± 0.004	0.838 ± 0.003	0.859 ± 0.001
ProGAN augmentation	0.918 ± 0.005	0.839 ± 0.000	0.855 ± 0.002
Weighted CE	0.915 ± 0.006	0.845 ± 0.004	0.854 ± 0.003
Focal loss	0.896 ± 0.014	0.854 ± 0.006	0.834 ± 0.012
OKO	0.892 ± 0.004	0.860 ± 0.006	0.832 ± 0.006
Balanced sampling	0.890 ± 0.004	0.848 ± 0.003	0.826 ± 0.002
CE + soft F1	0.876 ± 0.009	0.852 ± 0.006	0.813 ± 0.005
CE + soft MCC	0.874 ± 0.009	0.852 ± 0.006	0.811 ± 0.004

Table 20: PANDA native patch-feature test results (cancer vs. benign tiles) for validation-selected configurations, mean ± standard deviation across seeds.

remain within the broader CE overlap band, consistent with the interpretation that exposure corrections address only part of the error pattern when the representation already separates most classes. ProGAN augmentation does not deliver a clear advantage despite pixel-level FID values comparable to the source study (Ruiz-Casado et al., 2024); the decisive difference is that the source method trains its classifier end-to-end on pixels, whereas this benchmark feeds synthetic patches through a frozen encoder. Whether GAN-generated patches help under end-to-end encoder adaptation is left open as the most plausible route to a non-null synthetic-augmentation result. On WSI bags, RankMix and MDE-MIL both improve on plain MIL CE in every seed; the bottleneck includes instance selection and bag aggregation, so mixing informative subsequences (RankMix) and coupling natural and balanced branches with cross-expert consistency (MDE-MIL) can add value even when the underlying features are strong.

(b) Separability thesis: imbalance ratio is not the bottleneck. The three generality datasets show that the imbalance ratio does not predict whether mitigation helps. BRACS is mildly imbalanced (≈2.3:1) but hard (fine-grained breast-lesion subtyping entangled in the frozen feature space), so only OKO and CFAL show a measurable patch-level lift. CAMELYON16 is severely imbalanced (≈39:1) but easy (tumor and normal tiles are nearly separable in frozen Virchow2), so every method reaches near-ceiling discrimination and the unweighted baseline is competitive. PANDA provides the cleanest within-cohort control: two mildly imbalanced targets on the same biopsies diverge sharply in difficulty—binary cancer/benign detection (balanced accuracy ≈0.85) versus six-class ISUP grading (macro F1 ≈0.58)—while the no-mitigation baseline remains competitive in both regimes. This evidence makes imbalance

Method	Accuracy	Balanced accuracy	Macro F1
Balanced sampling	0.653 ± 0.028	0.588 ± 0.032	0.590 ± 0.033
Focal loss	0.654 ± 0.003	0.584 ± 0.016	0.583 ± 0.018
MDE-MIL	0.652 ± 0.029	0.581 ± 0.036	0.581 ± 0.040
RankMix	0.626 ± 0.014	0.571 ± 0.031	0.566 ± 0.034
MIL CE	0.639 ± 0.017	0.566 ± 0.016	0.565 ± 0.021
Weighted CE	0.639 ± 0.017	0.566 ± 0.016	0.565 ± 0.021
SC-MIL	0.619 ± 0.005	0.565 ± 0.025	0.559 ± 0.021

Table 21: PANDA native WSI-bag test results (slide-level six-class ISUP grading) for validation-selected configurations, mean ± standard deviation across seeds.

Method	ECE	ECE+TS	T
CE	0.046 ± 0.007	0.035 ± 0.003	2.733 ± 0.236
Weighted CE	0.045 ± 0.005	0.036 ± 0.002	2.703 ± 0.225
Focal loss	0.024 ± 0.000	0.031 ± 0.002	1.404 ± 0.126
CFAL	0.382 ± 0.004	0.011 ± 0.003	0.060 ± 0.001
CE + soft F1	0.103 ± 0.009	0.009 ± 0.001	6.418 ± 0.324
CE + soft MCC	0.101 ± 0.008	0.008 ± 0.003	5.491 ± 0.183
Balanced sampling	0.072 ± 0.005	0.019 ± 0.003	3.642 ± 0.184
OKO	0.028 ± 0.008	0.005 ± 0.002	0.822 ± 0.027
ProGAN augmentation	0.042 ± 0.007	0.039 ± 0.002	2.595 ± 0.343

Table 22: PANDA native patch-feature calibration: ECE before (raw) and after (TS) temperature scaling, with the fitted temperature T, mean ± standard deviation across seeds.

ratio alone an implausible driver: what governs the value of mitigation under a frozen-feature benchmark is class separability in the representation, not the head-to-tail ratio. This pattern is a direct caution against selecting or ranking imbalance-mitigation methods by imbalance ratio alone.

(c) ProGAN frozen-feature hypothesis. ProGAN augmentation is informative on BRACS and PANDA: the mild native imbalances on both datasets never create the tail-scarcity regime where synthetic minority augmentation helped on TCGA-UT at harsher constructed severities. On BRACS (BAcc 0.460 versus CE 0.451) and PANDA (macro F1 0.855 versus CE 0.859), ProGAN lands within between-seed noise of the CE baseline, neither helping nor hurting. This is consistent with the frozen-feature explanation for the TCGA-UT result rather than with a generator-quality failure: the generators reach reasonable pixel-level FID, but their outputs must pass through a frozen encoder that may map slightly off-manifold synthetic patches into embedding regions the head rarely sees for real data.

(d) Limitations. TCGA-UT patch labels come from pathologist-selected tumor regions; they are useful patch-level class labels but not dense annotations. The two benchmarks share case-disjoint splits but differ in prediction unit and per-slide evidence budget, so their result tables are non-transferable. The validation sweep varies one imbalance-specific control per method and does not exhaust all architecture or training-schedule choices. The benchmark is controlled rather than a full reproduction of each original paper: method sections document necessary adaptations to frozen Virchow2 features, and both regimes use frozen features rather than end-

Method	ECE	ECE+TS	T
MIL CE	0.237 ± 0.024	0.058 ± 0.013	3.410 ± 0.240
Weighted CE	0.237 ± 0.024	0.058 ± 0.013	3.410 ± 0.240
Focal loss	0.129 ± 0.022	0.062 ± 0.020	2.002 ± 0.166
Balanced sampling	0.241 ± 0.021	0.039 ± 0.009	3.723 ± 0.142
RankMix	0.144 ± 0.009	0.045 ± 0.020	1.530 ± 0.029
SC-MIL	0.252 ± 0.020	0.070 ± 0.015	3.452 ± 0.297
MDE-MIL	0.239 ± 0.035	0.097 ± 0.039	2.113 ± 0.080

Table 23: PANDA native WSI-bag calibration: ECE before and after temperature scaling, with the fitted temperature T. Columns as in Table 22.

Dataset	Regime	Best mitigation method	Baseline macro F1	Δ macro F1
TCGA-UT	Patch	CFAL	0.870	+0.008
TCGA-UT	WSI bag	MDE-MIL	0.843	+0.014
BRACS	Patch	CFAL	0.451	+0.039
BRACS	WSI bag	Weighted CE	0.508	−0.007
CAMELYON16	Patch	CFAL	0.958	+0.008
CAMELYON16	WSI bag	MDE-MIL / SC-MIL	0.965	+0.017
PANDA	Patch	CFAL	0.859	+0.007
PANDA	WSI bag	Balanced sampling	0.565	+0.025

Table 24: Cross-dataset mitigation headroom. Baseline is CE for patch regimes and MIL CE for WSI-bag regimes. Δ macro F1 is the best mitigation method's mean test macro F1 minus the no-mitigation baseline.

to-end encoder adaptation. Calibration results for BRACS and PANDA WSI-bag are affected by small test sets and fine-grained label sets rather than method-specific weaknesses.

6 Conclusion

This report benchmarked representative class-imbalance mitigation methods on four histopathology datasets spanning the difficulty–imbalance plane, using frozen Virchow2 features, bounded per-slide evidence budgets, and patient-disjoint splits throughout. On TCGA-UT, the anchor dataset with the richest analysis, CFAL reaches the highest patch-level mean macro F1; OKO set learning reaches the highest balanced accuracy while achieving the best raw NLL and Brier among patch methods; RankMix has the highest WSI-bag balanced accuracy and MDE-MIL the highest WSI-bag macro F1. The three generality datasets show that these results do not generalize uniformly: on BRACS only OKO and CFAL visibly separate from the CE baseline; on CAMELYON16 every method reaches near-ceiling and the unweighted baseline is competitive; and on PANDA both mildly imbalanced regimes are won by the no-mitigation baseline within between-seed noise. The practical implication is that the value of imbalance mitigation under a frozen-feature benchmark is governed by class separability in the representation, not by the head-to-tail ratio. Imbalance mitigation should therefore be evaluated in the native input regime and alongside classwise recall and calibration, not only by aggregate discrimination, and method rankings should not be transferred across datasets on the basis of imbalance ratio alone.

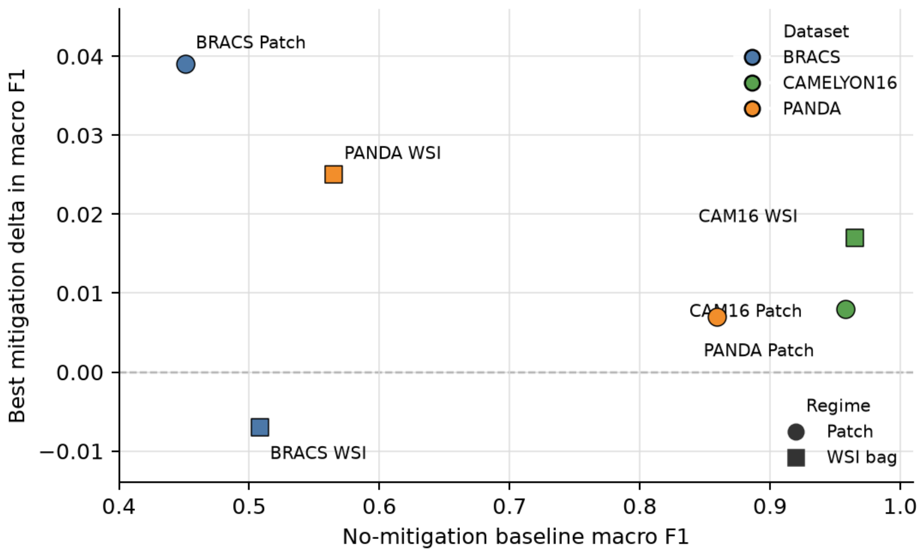

Figure 2: Relationship between no-mitigation baseline macro F1 and mitigation headroom across all dataset–regime pairs. High-baseline regimes leave little room for mitigation; lower-baseline regimes show the only material positive deltas, although BRACS WSI-bag shows that low baseline performance alone does not guarantee an effective mitigation method.

A Additional TCGA-UT Result Tables and Figures

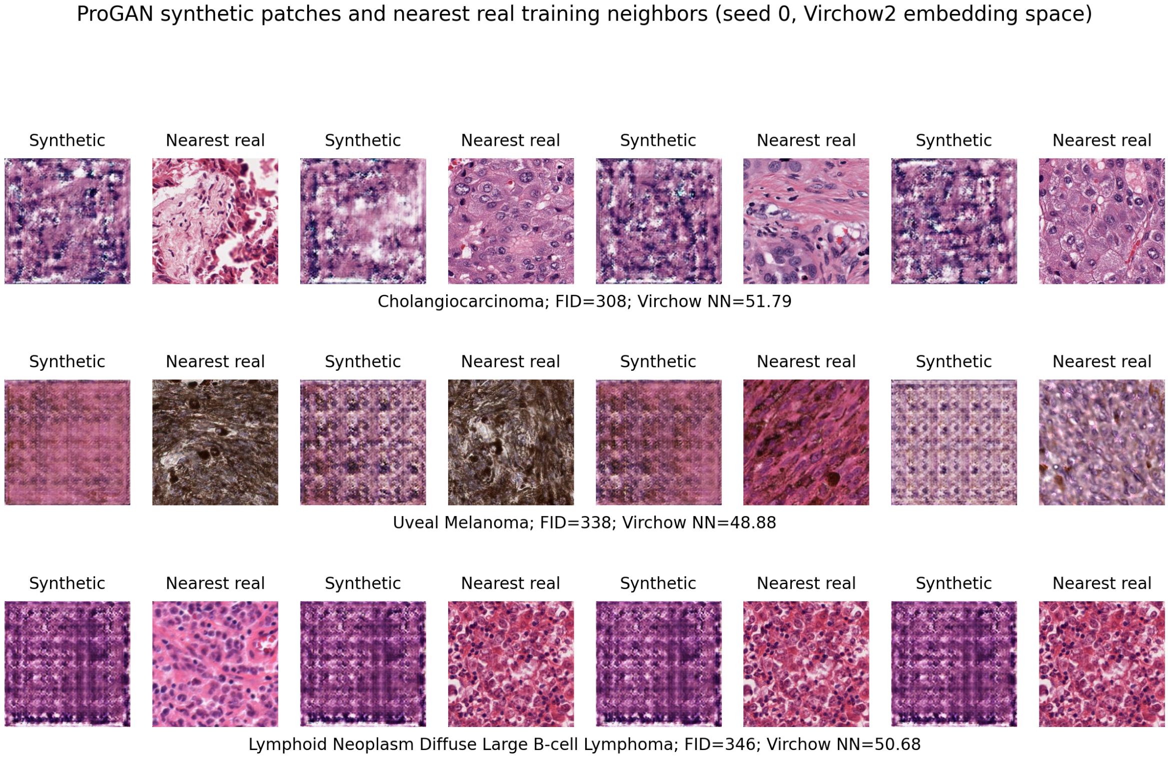

Figure 3: ProGAN synthetic patches (left column of each pair) and their nearest real training neighbors in Virchow2 embedding space (right column) for three tail classes in seed 0. Visual quality is class-dependent. Per-class Inception FID values are comparable to those reported by Ruiz-Casado et al. (2024) for ProGAN on a binary breast-histology task; the frozen-feature pipeline—not generator quality—is the proposed explanation for the null downstream result.

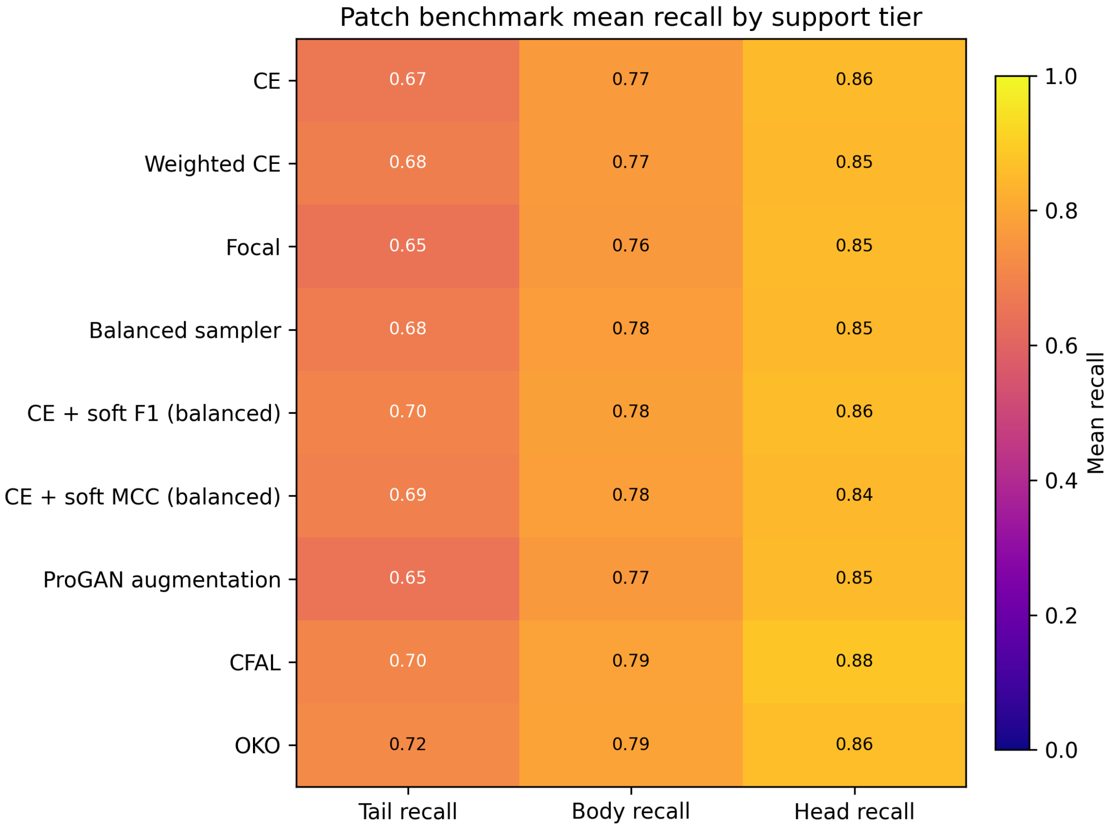

Figure 4: TCGA-UT patch-level mean recall by support tier on the held-out test split. The heatmap exposes the head–body–tail variation hidden by aggregate accuracy.

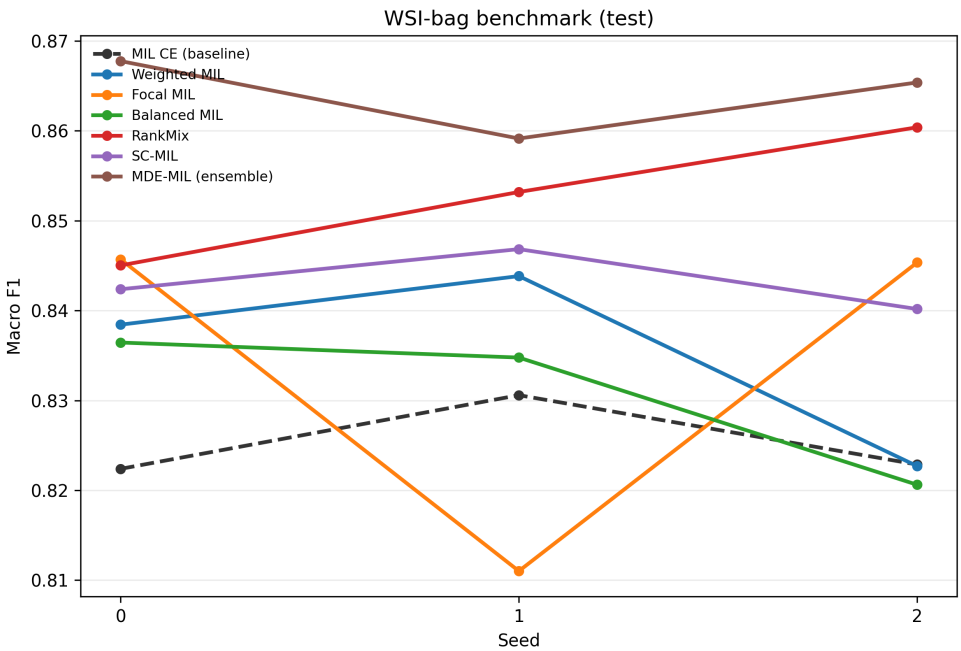

Figure 5: Validation-selected TCGA-UT WSI-bag macro F1 on the held-out test split by random seed. The dashed line marks plain MIL CE. The pronounced focal-MIL dip on seed 1 is a single-seed training instability; its averaged result reflects one unstable run rather than a consistent deficit.

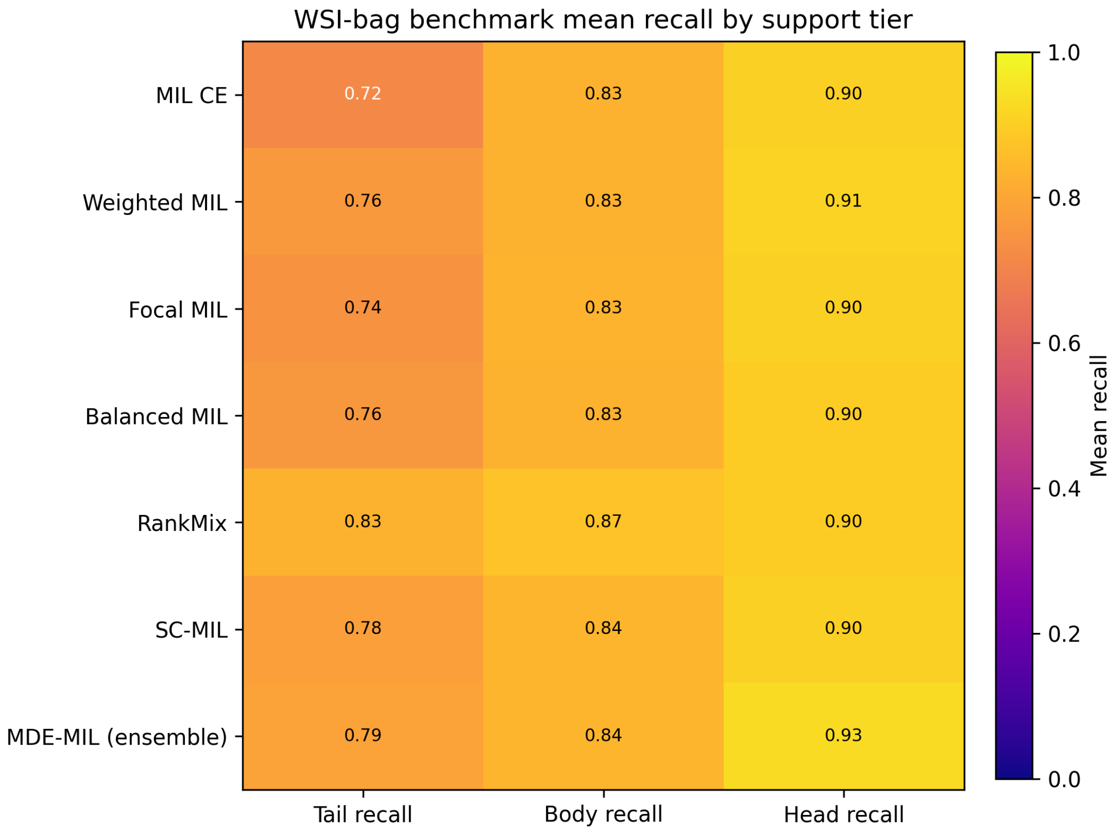

Figure 6: TCGA-UT WSI-bag classwise recall on the held-out test split. The spread across classes shows why aggregate accuracy alone is insufficient for the imbalanced setting.

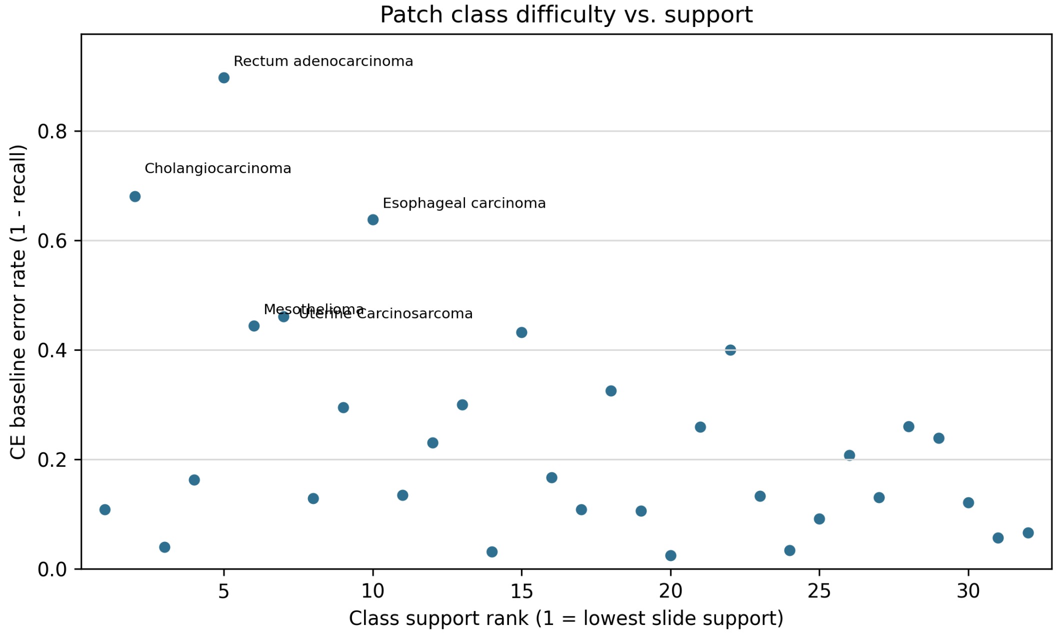

Figure 7: TCGA-UT patch CE-baseline class error rate versus full-dataset class-support rank. The hardest patch classes are not simply ordered by rarity, indicating that exposure-based corrections address only part of the error pattern.

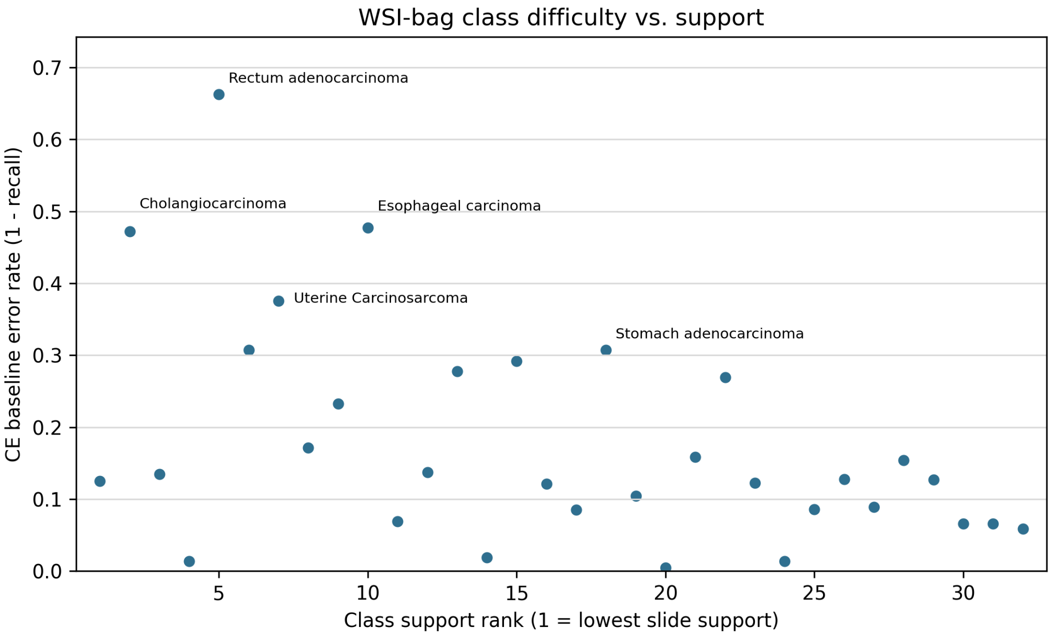

Figure 8: TCGA-UT WSI-bag CE-baseline class error rate versus full-dataset class-support rank. The same high-error classes are visible at slide level, showing that class difficulty is not reducible to class frequency.

Comparison	Δ Macro F1	Δ Balanced accuracy
CE + soft MCC − CE	-0.003 ± 0.008 [$+0.005, -0.000, -0.014$]	$+0.009 \pm 0.007$ [$+0.015, +0.012, -0.000$]
CE + soft F1 − CE	$+0.007 \pm 0.010$ [$+0.014, +0.014, -0.007$]	$+0.018 \pm 0.008$ [$+0.023, +0.023, +0.007$]
OKO − CE	$+0.015 \pm 0.010$ [$+0.024, +0.021, +0.001$]	$+0.028 \pm 0.012$ [$+0.038, +0.033, +0.011$]
CFAL − CE	$+0.025 \pm 0.005$ [$+0.030, +0.027, +0.019$]	$+0.026 \pm 0.004$ [$+0.030, +0.027, +0.020$]
RankMix − MIL CE	$+0.028 \pm 0.010$ [$+0.023, +0.041, +0.020$]	$+0.041 \pm 0.010$ [$+0.038, +0.055, +0.031$]
MDE-MIL − MIL CE	$+0.039 \pm 0.016$ [$+0.055, +0.043, +0.018$]	$+0.026 \pm 0.016$ [$+0.041, +0.034, +0.003$]

Table 25: TCGA-UT paired validation-selected test-split differences against the within-regime CE baseline on seeds 0–2. Each cell reports mean ± SD across seeds; bracketed values list seed-ordered paired deltas. No formal p-values are computed.

Regime	Method	NLL	Brier	ECE
Patch	CFAL	2.850 ± 0.011	0.914 ± 0.001	0.787 ± 0.008
Patch	OKO	0.756 ± 0.040	0.270 ± 0.015	0.158 ± 0.013
Patch	CE + soft F1 (balanced)	3.038 ± 0.137	0.316 ± 0.008	0.155 ± 0.004
Patch	Weighted CE	1.965 ± 0.115	0.314 ± 0.016	0.147 ± 0.007
Patch	CE	1.976 ± 0.149	0.312 ± 0.018	0.147 ± 0.009
Patch	Balanced sampler	1.967 ± 0.139	0.313 ± 0.019	0.147 ± 0.009
Patch	CE + soft MCC (balanced)	2.870 ± 0.153	0.332 ± 0.017	0.162 ± 0.008
Patch	ProGAN augmentation	2.511 ± 0.093	0.327 ± 0.007	0.157 ± 0.004
Patch	Focal	1.370 ± 0.081	0.298 ± 0.014	0.131 ± 0.006
WSI bag	MDE-MIL (ensemble)	0.487 ± 0.005	0.166 ± 0.002	0.036 ± 0.004
WSI bag	RankMix	0.584 ± 0.029	0.170 ± 0.006	0.049 ± 0.001
WSI bag	SC-MIL	1.024 ± 0.145	0.204 ± 0.020	0.092 ± 0.009
WSI bag	Weighted MIL	1.224 ± 0.144	0.212 ± 0.012	0.099 ± 0.006
WSI bag	Balanced MIL	1.264 ± 0.165	0.223 ± 0.010	0.102 ± 0.005
WSI bag	MIL CE	1.283 ± 0.080	0.227 ± 0.024	0.106 ± 0.012
WSI bag	Focal MIL	1.117 ± 0.153	0.221 ± 0.014	0.098 ± 0.009

Table 26: TCGA-UT held-out test raw calibration metrics for validation-selected configurations over three seeds. Lower is better for NLL, Brier, and ECE. Values are not comparable across regimes. Table 10 gives the post-hoc counterpart.

B Terminology

Term	Meaning in this study
Slide / WSI	Diagnostic whole-slide image carrying one label.
Case / patient	TCGA participant identified by the first three fields of the TCGA slide barcode (e.g. `TCGA-OR-A5J1`); or the equivalent patient identifier in BRACS and PANDA.
Patch	256×256 H&E crop from a selected region; patch labels inherit the slide-level class.
Patch embedding	Frozen Virchow2 vector for one controlled patch; input to the patch-level classifier head.
Feature chunk	Frozen Virchow2 feature representation associated with a slide or slide subset.
WSI bag	Set of patch-feature instances for one slide, consumed by the MIL model.
Head/body/tail class	Fixed support tier from the full TCGA-UT slide distribution (8 each for head and tail; 16 body).
Case-disjoint split	Split rule preventing any participant's slides, patches, or features from crossing train, validation, and test.
Evidence budget	Per-slide cap on the number of patch or bag instances; dataset-specific but follows a common data-driven rule (Table 1).

Table 27: Terminology used across all four dataset benchmarks.

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